

Muhammad Bilal Khan

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EDUCATION

- University of Waterloo** Waterloo, ON, Canada
 - Bachelor of Mathematics in Statistics, Computational Mathematics.* Sept. 2020 – June 2025
 - GPA: 3.8, President's Scholarship of \$5000, Graduated with Honours.

SKILLS

- Languages:** Python, C++, Javascript, TypeScript, SQL, Java, R.
- Tools:** AWS, GCP, Langchain, LangGraph, ElasticSearch, Nginx, Kafka, MySQL, FastAPI, MongoDB, Airflow, Kubernetes, Prometheus, Grafana, Jenkins, Spark, PyTorch, Tensorflow, CUDA, XLA, MLIR.

EXPERIENCE

- Ericsson** Ottawa, ON, Canada
 - Software Engineer* April 2024 - Present
 - Cut manual provisioning by **50%** saving est. **\$40,000**/year by shipping a LangGraph-based modular agentic tool for automated orchestration of **200+** CloudRAN simulation environments. Used by **8** teams across Ottawa lab.
 - Powering agent's RAG + CoT architecture by orchestrating Spark/Kafka data pipeline. Ingesting **15TB/year** from **100+** distributed 5G devices using HTTPS/gRPC requests.
 - Enabled **8+** lab teams with real-time data via public REST API (FastAPI) for reading from MySQL (metrics) and Elasticsearch (logs). Published with OpenAPI docs, monitoring to ensure reliable, low-latency access at scale.
 - Deployed on Google Kubernetes Engine with NGINX ingress for robust MLOps scalability, with Jenkins and Airflow CI/CD for model versioning and continuous training. Grafana dashboard for performance profiling.
 - Trained Informer-based forecasting models on GCP Vertex AI on real-time metrics to assist agentic flow and drive predictive lab orchestration. Evaluated agents with RAGAS, A/B Testing, and forecasts on labelled usage data.
- Vosyn** Toronto, ON, Canada
 - ML Engineer Intern* June 2023 - Dec 2023
 - Implemented CUDA kernels to accelerate ASR/TTS model inference on A100 GPUs by **15%**, performing sparse matrix multiplication, speech waveform convolutions and parallel inference.
 - Shipped models trained on AWS SageMaker and implemented gRPC endpoints for scalable model serving and end-to-end deployment. Presented to investors in funding rounds; contributed to **\$36.4M** startup valuation.
 - Finetuned Open AI Whisper model, lowered word error rate by **10%** in **5** low-data languages for content translation. Involved QLoRA, data augmentation on public datasets, colearning and multimodal data fusion.
- University of Waterloo** Waterloo, ON, Canada
 - ML Researcher* Sept. 2023 - March 2024
 - Co-authored a publication with prof. Zelalem Negeri and NSERC's **\$10,000** grant, pioneering predictive diagnostics in healthcare; cited **twice**. Presented paper to **est. 200** international academics at **JRC 2024**.
 - Designed novel Monte Carlo estimation for causal ML models in Pytorch. Tested each algorithm's convergence, estimation paths and errors using meta-analyses data from Cochrane Database.
 - Recommended future directions of research by identifying relationships between method choice and data characteristics. Paper: <https://shorturl.at/y8TkY>

PROJECTS

- 1st Place at Cohere x OpenSesame:** Won hackathon by developing Syntaxtual, an agentic VS Code plugin for live code review with repository-level context, allowing for conversational issue resolution.
- 2nd Place at MAG Aerospace Wildfire Hackathon:** Predicted **10 days** of future fire growth for Alberta's **Jasper Wildfire** with topography, fire fuel, and fire hotspot raster data. **Published** in UW Data & AI Institute Newsletter.
- MedChat at Waterloo.AI:** Built LLM chatbot finetuned on PubMed literature at **largest undergraduate AI research** hub in Canada. Achieves **75%** accuracy on symptoms/diagnosis benchmark with agentic RAG, Llama 3.
- Lead Developer at Global Spark, BCG:** Mentoring, hosting workshops on LLMs, Computer Vision. Developed NLP-based ATS system for HackTheGlobe applications by finetuning Cohere R7B with AutoML.